

Paper Practice 8.3**Negative Exponents**

WORKED KEY – Simplify each expression. Assume no denominator equals zero. Write every answer with positive exponents only.

1. Simplify. (Zero Exponents)

a. n^0 ANSWER: 1

b. $(-387)^0$ ANSWER: 1

c. $5x^0$ ANSWER: 5 (only x is raised to the zero power, not the 5)

2. Write each expression with positive exponents only. (Negative Exponents)

a. n^{-5} ANSWER: $1/n^5$

b. 3^{-2} ANSWER: $1/9$

c. $(-2)^{-3}$ ANSWER: $-1/8$

3. Simplify. Write with positive exponents. (Negative Exponents)

a. $a^4b^{-2}c^{-3}$ ANSWER: $a^4/(b^2c^3)$

b. m^2p^{-3} ANSWER: m^2/p^3

4. Simplify. (Quotients)

a. a^5b^3/a^2 ANSWER: a^3b^3

b. $6u^3v^7w^0/(2uv^4)$ ANSWER: $3u^2v^3$ ($w^0 = 1$)

5. Simplify. (Quotients)

a. $5q^4r^5t^3/(11rt^3)$ ANSWER: $5q^4r^4/11$ ($t^{3-3} = t^0 = 1$)

b. $3g^4h^3/(-36g^4hj^2)$ ANSWER: $-h^2j^2/(12g^8)$

6. Simplify. (Zero Exponents)

a. $(-8m^3np^6/(9k^3mn^2))^0$ ANSWER: 1 – the whole quantity is raised to the zero power.

b. $4x^0 + (7y)^0$ ANSWER: $4 + 1 = 5$

7. Simplify. Write with positive exponents. (Mixed)

a. $-c^3f^2d^{-1}e^4$ ANSWER: $-c^3e^4/(f^2d)$

b. $(2x^{-3})(5x^7)$ ANSWER: $10x^4$

8. Evaluate. (Evaluate)

a. 2^{-4} ANSWER: $1/16$

b. $(\frac{1}{3})^{-2}$ ANSWER: 9

c. $5^0 + 5^{-1}$ ANSWER: $1 + 1/5 = 6/5$

9. Simplify $(3a^2b^{-1})^2$. Write with positive exponents. (Power of a Power) ANSWER:
 $3^{-2}a^{-4}b^2 = b^2/(9a^4)$

10. Challenge – simplify $-8r^0t^{-3}v^{-10}/(2rt^2v^{-5})$.

(Challenge) ANSWER: $-4r^{-1}t^{-5}v^{-5} = -4/(rt^5v^5)$

11. Error Analysis – Gus wrote $x^{-3} = -x^3$. Identify his mistake and write the correct form. (Error Analysis) ANSWER: A negative exponent does not make the value negative – it means the reciprocal. $x^{-3} = 1/x^3$.

12. Explain, using the quotient rule, why any nonzero number raised to the zero power equals 1. (Explain) ANSWER: $a^n/a^n = 1$, and by the quotient rule it also equals $a^{n-n} = a^0$. So $a^0 = 1$.